

Causal inference with semi-competing risks, and beyond

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October 23, 2026

Outline

Motivation

Formalization of Causal Semi-Competing Risks

Natural Effects

Application to Stem Cell Transplantation Data

Concluding Remarks

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Formalization of Causal Semi-Competing Risks

Natural Effects

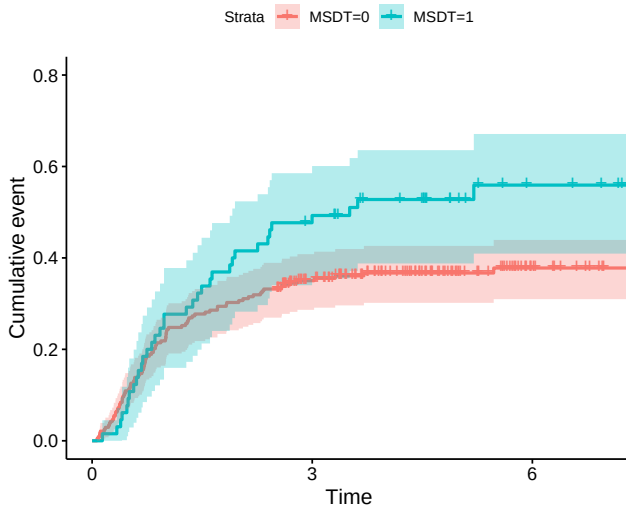
Application to Stem Cell Transplantation Data

Concluding Remarks

Motivating Data

- Stem cell transplantation (SCT) is a commonly adopted approach to cure leukemia.
- Human leukocyte antigen matched sibling donor transplantation (MSDT) has long been considered as the first choice of transplantation because of lower transplant-related mortality.
- In recent years, some benefits of haploidentical stem cell transplantation (haplo-SCT) have been found in that haplo-SCT results in lower relapse rate.
- What is the mechanism of transplant modality on overall mortality (transplant-related mortality, relapse-related mortality)?

Motivating Data: Mortality



Question

- The question is about mediation analysis.
- The mediator and outcome are “time to event”, subject to censoring.
- We want to know
 - (1) the treatment effect delivered *directly* to the terminal event and
 - (2) the treatment effect delivered *indirectly* through (*mediated* by) the intermediate event to the terminal event.
- How to define, identify, estimate, infer and interpret the causal effect?

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Mediation Analysis at Varying Time Points

- For time-to-event data, the mediator and outcome are processes.
- The mediator (intermediate event) process $N_1(t) \in \{0, 1\}$: whether the intermediate event has occurred at t .
- The outcome (terminal event) process $N_2(t) \in \{0, 1\}$: whether the terminal event has occurred at t .

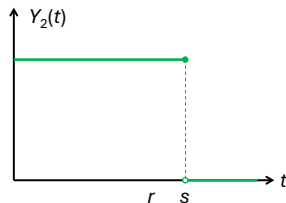
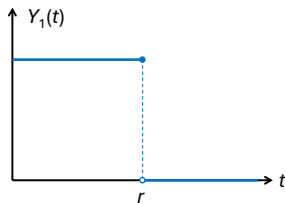
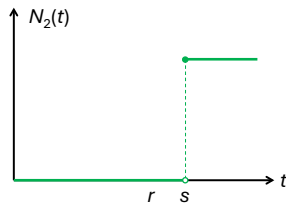
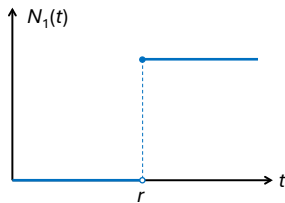
$N_1(t^-)$	$N_2(t)$	Description at time t
0	0	No intermediate event, no terminal event
0	1	No intermediate event, a terminal event
1	0	An intermediate event, no terminal event
1	1	An intermediate event, a terminal event

Semi-Competing Risks

- *Semi-competing risks* refer to the phenomenon that the terminal event can truncate the non-terminal event but not vice versa (Fine et al, 2001; Huang, 2021).
- The terminal event and non-terminal events have a competing nature, but their roles are not the same.
- Related to the truncation-by-death problem.

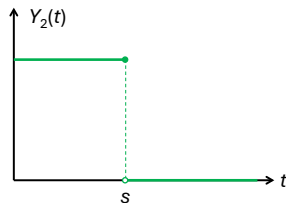
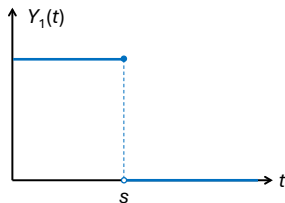
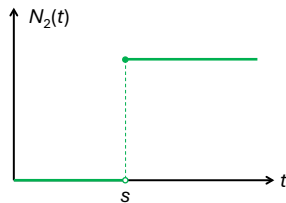
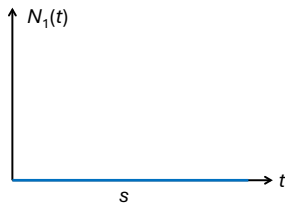
Event Processes and At-Risk Processes

- Intermediate event occurs first.



Event Processes and At-Risk Processes

- Intermediate event does not occur.



Formal Definition: Jumps

- Let $d\tilde{N}_1(t; z_1)$ be the jump of potential event counting process for the intermediate event during $[t, t + dt)$ when the treatment is set at z_1 .
- Let $d\tilde{N}_2(t; z_2, n_1)$ be the jump of potential event counting process for the terminal event during $[t, t + dt)$ when the treatment is set at z_2 and the counting process for the intermediate event at t^- is set at n_1 .

Formal Definition: Well Definedness

- **Markovness:** the hazard of jumps only relies on the current status, not the history,

$$\begin{aligned} &P(d\tilde{N}_2(t; z_2, n_1) = 1 \mid \tilde{N}_2(t^-; z_2, \tilde{n}_1(\cdot)) = 0, \tilde{n}_1(t^-) = n_1, \tilde{n}_1(s^-) = n_1^*) \\ &= P(d\tilde{N}_2(t; z_2, n_1) = 1 \mid \tilde{N}_2(t^-; z_2, \tilde{n}_1(\cdot)) = 0, \tilde{n}_1(t^-) = n_1), \quad s < t. \end{aligned}$$

- Markovness ensures that the notation $d\tilde{N}_2(t; z_2, n_1)$ is well defined.
- We can manipulate the treatment z_2 , or we can manipulate the status of the intermediate event n_1 , in order to generate a potential jump $d\tilde{N}_2(t; z_2, n_1)$.

Formal Definition: Processes

- We integrate the jumps to obtain potential counting processes.
- The potential event counting process for the intermediate event

$$\tilde{N}_1(t; \mathbf{z}_1) = \int_0^t d\tilde{N}_1(s; \mathbf{z}_1).$$

- Given the full intermediate event process $\tilde{n}_1(\cdot)$, the potential event counting process for the terminal event

$$\tilde{N}_2(t; \mathbf{z}_2, \tilde{n}_1(\cdot)) = \int_0^t d\tilde{N}_2(s; \mathbf{z}_2, \tilde{n}_1(s^-)).$$

Consistency and Cross-Worlds

- The potential time to intermediate event $\tilde{T}_1(\mathbf{z}_1)$ is the time that $\tilde{N}_1(t; \mathbf{z}_1)$ jumps.
- Substituting $\tilde{n}_1(\cdot)$ with $\tilde{N}_1(\cdot; \mathbf{z}_1)$, the potential time to terminal event $\tilde{T}_2(\mathbf{z}_1, \mathbf{z}_2)$ is the time that $\tilde{N}_2(t; \mathbf{z}_2, \tilde{N}_1(\cdot; \mathbf{z}_1))$ jumps.
- **Causal consistency:**

$$\tilde{N}_1(t) = \tilde{N}_1(t; Z), \quad \tilde{N}_2(t) = \tilde{N}_2(t; Z, \tilde{N}_1(\cdot; Z)), \quad t \in [0, t^*].$$

- Under causal consistency, $\tilde{T}_1 = \tilde{T}_1(Z)$ and $\tilde{T}_2 = \tilde{T}_2(Z, Z)$.

Estimand

- The *counterfactual cumulative incidence* of the terminal event

$$F(t; \mathbf{z}_1, \mathbf{z}_2) = P(\tilde{T}_2(\mathbf{z}_1, \mathbf{z}_2) \leq t) = P(\tilde{N}_2(t; \mathbf{z}_2, \tilde{N}_1(\cdot; \mathbf{z}_1)) = 1)$$

is of primary interest.

- To identify $F(t; \mathbf{z}_1, \mathbf{z}_2)$, it is equivalent to identify the hazard

$$d\Lambda(t; \mathbf{z}_1, \mathbf{z}_2) = d \log\{1 - F(t; \mathbf{z}_1, \mathbf{z}_2)\}.$$

- Total treatment effect: $F(t; 1, 1) - F(t; 0, 0)$.
- Natural direct effect: $F(t; 0, 1) - F(t; 0, 0)$.
- Natural indirect effect: $F(t; 1, 1) - F(t; 0, 1)$.

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Assumptions for Identification

- **Ignorability:** the treatment is randomized,

$$\{d\tilde{N}_1(t; z_1), d\tilde{N}_2(t; z_2, n_1) : 0 \leq t \leq t^*\} \perp\!\!\!\perp Z.$$

- **Random censoring:**

$$\{\tilde{N}_1(t; z), \tilde{N}_2(t; z, \tilde{N}_1(\cdot; z)) : 0 \leq t \leq t^*\} \perp\!\!\!\perp C \mid Z.$$

- You may consider C as a potential outcome since C is a post-treatment value, in which case ignorability should be assumed for $C(z)$.

Assumptions for Identification

- **Positivity 1:**

$$P(\tilde{N}_1(t; z_1) = n_1, \tilde{N}_2(t; z_2, \tilde{N}_1(\cdot; z_1)) = 0) > 0$$
$$\Rightarrow P(Z = z \mid \tilde{N}_1(t; z) = n_1, \tilde{N}_2(t; z, \tilde{N}_1(\cdot; z)) = 0) > 0, \quad z \in \{0, 1\},$$

$$z_1, z_2, n_1 \in \{0, 1\}, t \in [0, t^*].$$

- **Positivity 2:**

$$P(C > t^* \mid Z) > 0.$$

Observed Processes

- The event times $T_1 = \tilde{T}_1 \wedge C$ and $T_2 = \tilde{T}_2 \wedge C$.
- The censoring indicators $\delta_1 = I\{\tilde{T}_1 \leq C\}$ and $\delta_2 = I\{\tilde{T}_2 \leq C\}$.
- The observed counting process for the intermediate event $N_*(t; z) = I\{T_1 \leq t, T_2 \geq T_1, \delta_1 = 1, Z = z\}$.
- The observed at-risk process for the intermediate event $Y_*(t; z) = I\{T_1 \geq t, T_2 \geq t, Z = z\}$.
- Hereafter we use the subscript “*” to represent the intermediate event, “1” to represent the terminal event with prior intermediate event (direct terminal event), and “0” to represent the terminal event without prior intermediate event (indirect terminal event).

Observed Processes

- The observed counting process for the direct terminal event
 $N_0(t; z) = I\{T_2 \leq t, T_1 \geq T_2, \delta_2 = 1, Z = z\}.$
- The observed at-risk process for the direct terminal event
 $Y_0(t; z) = I\{T_2 \geq t, T_1 \geq t, Z = z\}.$
- The observed counting process for the indirect terminal event
 $N_1(t; z) = I\{T_2 \leq t, T_1 < t, \delta_2 = 1, Z = z\}.$
- The observed at-risk process for the indirect terminal event
 $Y_1(t; z) = I\{T_2 \geq t, T_1 < t, Z = z\}.$
- $Y_*(t; z) = Y_0(t; z)$ because the intermediate event and the direct terminal event are a pair of competing events, sharing the same at-risk set.

Cause-Specific Hazards

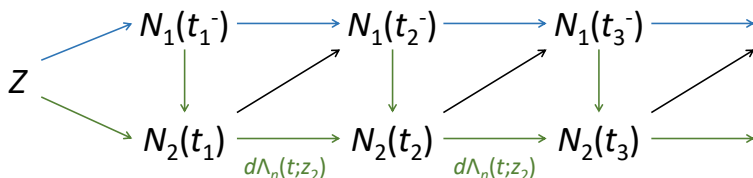
- Define the cause-specific hazard of the terminal event

$$d\Lambda_{n_1}(t; \mathbf{z}_1, \mathbf{z}_2) \\ = P(d\tilde{N}_2(t; \mathbf{z}_2, n_1) = 1 \mid \tilde{N}_1(t^-; \mathbf{z}_1) = n_1, \tilde{N}_2(t^-; \mathbf{z}_2, \tilde{N}_1(\mathbf{z}_1)) = 0).$$

- It involves cross-world values. How to identify the cause-specific hazard?
- Recall the assumptions we have made: Markovness, consistency, ignorability, random censoring, and positivity.
- Compared to mediation analysis, what assumption is lost? — Sequential ignorability!

Competing Events in the Cross-World

- Note that $\tilde{N}_1(\cdot)$ and $\tilde{N}_2(\cdot)$ are competing. In the cross-world (z_1, z_2) , $\tilde{N}_2(\cdot)$ relies on z_1 and $\tilde{N}_1(\cdot)$ relies on z_2 .
- We need two sequential ignorability assumptions, one for $\tilde{N}_2(\cdot)$ and the other for $\tilde{N}_1(\cdot)$.



Sequential Ignorability, Part 1

- **Sequential ignorability (1):**

$$\begin{aligned} &P(d\tilde{N}_2(t; \mathbf{z}_2, n_1) = 1 \mid Z = \mathbf{z}_2, \tilde{N}_1(t^-; \mathbf{z}_1) = n_1, \tilde{N}_2(t^-; \mathbf{z}_2, \tilde{N}_1(\cdot; \mathbf{z}_1)) = 0) \\ &= P(d\tilde{N}_2(t; \mathbf{z}_2, n_1) = 1 \mid Z = \mathbf{z}_2, \tilde{N}_1(t^-) = n_1, \tilde{N}_2(t^-) = 0). \end{aligned}$$

- At time t , given the history, the “potential outcome” $d\tilde{N}_2(t; Z, n_1)$ is independent of the “treatment” $\tilde{N}_1(t^-; \mathbf{z}_1)$:

$$d\tilde{N}_2(t; \mathbf{z}_2, n_1) \perp\!\!\!\perp \tilde{N}_1(t^-; \mathbf{z}_1) \mid Z = \mathbf{z}_2, \tilde{N}_2(t^-; \mathbf{z}_2, n_1) = 0.$$

- No cross-world reliance of $d\tilde{N}_2(t; \mathbf{z}_2, n_1)$ on $\mathbf{z}_1$.

Sequential Ignorability, Part 1

- The cause-specific hazard (of the terminal event) is identifiable

$$\begin{aligned} & d\Lambda_{n_1}(t; \mathbf{z}_1, \mathbf{z}_2) \\ &= P(d\tilde{N}_2(t; \mathbf{z}_2, n_1) = 1 \mid \tilde{N}_1(t^-; \mathbf{z}_1) = n_1, \tilde{N}_2(t^-; \mathbf{z}_2, \tilde{N}_1(\cdot; \mathbf{z}_1)) = 0) \\ &= P(d\tilde{N}_2(t; \mathbf{z}_2, n_1) = 1 \mid Z = \mathbf{z}_2, \tilde{N}_1(t^-) = n_1, \tilde{N}_2(t^-) = 0) \end{aligned}$$

and can be estimated by Nelson–Aalen estimators,

$$d\hat{\Lambda}_{n_1}(t; \mathbf{z}_2) = \frac{I\{\bar{Y}_{n_1}(t; \mathbf{z}_2) > 0\}}{\bar{Y}_{n_1}(t; \mathbf{z}_2)} d\bar{N}_{n_1}(t; \mathbf{z}_2).$$

Sequential Ignorability, Part 2

- **Sequential ignorability (2):**

$$\begin{aligned} &P(d\tilde{N}_1(t; \mathbf{z}_1) = 1 \mid Z = \mathbf{z}_1, \tilde{N}_1(t^-; \mathbf{z}_1) = 0, \tilde{N}_2(t^-; \mathbf{z}_2, \tilde{N}_1(\cdot; \mathbf{z}_1)) = 0) \\ &= P(d\tilde{N}_1(t; \mathbf{z}_1) = 1 \mid Z = \mathbf{z}_1, \tilde{N}_1(t^-) = 0, \tilde{N}_2(t^-) = 0). \end{aligned}$$

- At time t , given the history, the “potential outcome” $d\tilde{N}_1(t; Z)$ is independent of the “treatment” $\tilde{N}_2(t^-; \mathbf{z}_2, \tilde{N}_1(\cdot; \mathbf{z}_1))$:

$$d\tilde{N}_1(t; \mathbf{z}_1) \perp\!\!\!\perp \tilde{N}_2(t^-; \mathbf{z}_2, \tilde{n}_1(\cdot)) \mid Z = \mathbf{z}_1, \tilde{N}_1(t^-; \mathbf{z}_1) = \tilde{n}_1(t^-).$$

- No cross-world reliance of $d\tilde{N}_1(t; \mathbf{z}_1)$ on $\mathbf{z}_2$.

Sequential Ignorability, Part 2

- The hazard of the intermediate event is identifiable

$$\begin{aligned} & d\Lambda_*(t; \mathbf{z}_1, \mathbf{z}_2) \\ &= P(d\tilde{N}_1(t; \mathbf{z}_1) = 1 \mid \tilde{N}_1(t^-; \mathbf{z}_1) = 0, \tilde{N}_2(t^-; \mathbf{z}_2, \tilde{N}_1(\cdot; \mathbf{z}_1)) = 0) \\ &= P(d\tilde{N}_1(t; \mathbf{z}_1) = 1 \mid Z = \mathbf{z}_1, \tilde{N}_1(t^-) = 0, \tilde{N}_2(t^-) = 0) \end{aligned}$$

and can be estimated by Nelson–Aalen estimators,

$$d\hat{\Lambda}_*(t; \mathbf{z}_1) = \frac{I\{\bar{Y}_*(t; \mathbf{z}_1) > 0\}}{\bar{Y}_*(t; \mathbf{z}_1)} dN_*(t; \mathbf{z}_1).$$

Subdistribution

- We partition the terminal event into a direct event (which does not have a history of intermediate event) and an indirect event (which has a history of intermediate event).
- The cumulative incidence of the terminal event consists of two subdistributions,

$$F(t; z_1, z_2) = F_0(t; z_1, z_2) + F_1(t; z_1, z_2),$$

where

$$\begin{aligned} F_0(t; z_1, z_2) &= P(\tilde{N}_2(t; z_2, \tilde{N}_1(\cdot; z_1)) = 1, \tilde{N}_1(t; z_1) = 0) \\ &= \int_0^t \exp\{-\Lambda_*(s^-; z_1) - \Lambda_0(s^-; z_2)\} d\Lambda_0(s; z_2), \\ F_1(t; z_1, z_2) &= P(\tilde{N}_2(t; z_2, \tilde{N}_1(\cdot; z_1)) = 1, \tilde{N}_1(t; z_1) = 1) \\ &= \int_0^t \exp\{-\Lambda_*(s^-; z_1) - \Lambda_0(s^-; z_2)\} [1 - \exp\{-\Lambda_1(t; z_2) + \Lambda_1(s; z_2)\}] d\Lambda_*(s; z_1). \end{aligned}$$

Interpretation: Controlling the Hazard

- The natural direct effect measures the treatment effect on the cumulative incidence of terminal event via changing the cause-specific hazards of terminal events while controlling the *hazard* of intermediate events.
- The natural indirect effect measures the treatment effect on the cumulative incidence of terminal event via changing the *hazard* of intermediate events while controlling the cause-specific hazards of terminal events.
- Nevertheless, since we have no information of the cross-world, whether the assumption is appropriate is worthy of extensive discussion.

An Alternative to Sequential Ignorability (2)

- **Sequential ignorability (2)**, prevalence version:

$$\begin{aligned} &P(\tilde{N}_1(t; \mathbf{z}_1) = 1 \mid Z = \mathbf{z}_1, \tilde{N}_1(t^-; \mathbf{z}_1) = 0, \tilde{N}_2(t^-; \mathbf{z}_2, \tilde{N}_1(\cdot; \mathbf{z}_1)) = 0) \\ &= P(\tilde{N}_1(t; \mathbf{z}_1) = 1 \mid Z = \mathbf{z}_1, \tilde{N}_1(t^-) = 0, \tilde{N}_2(t^-) = 0). \end{aligned}$$

- It assumes $\tilde{N}_1(\cdot; \mathbf{z}_1)$ is completely independent of $\tilde{N}_2(t^-; \mathbf{z}_2, \tilde{n}_1(\cdot))$:

$$\tilde{N}_1(\cdot; \mathbf{z}_1) \perp\!\!\!\perp \tilde{N}_2(t^-; \mathbf{z}_2, \tilde{n}_1(\cdot)) \mid Z = \mathbf{z}_1.$$

- No cross-world reliance of $\tilde{N}_1(t; \mathbf{z}_1)$ on $\mathbf{z}_2$.
- This independency fails, and should not be assumed, for $\mathbf{z}_1 = \mathbf{z}_2$.

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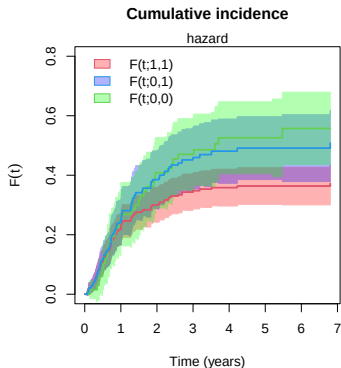
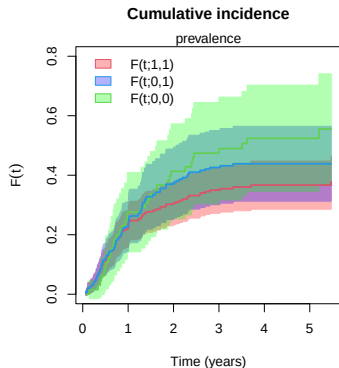
Concluding Remarks

Leukemia Data

- Allogeneic stem cell transplantation is a commonly adopted approach to cure acute lymphoblastic leukemia.
- Human leukocyte antigen matched sibling donor transplantation (MSDT) has long been considered as the first choice of transplantation because of lower transplant-related mortality.
- In recent years, some benefits of haploidentical stem cell transplantation (haplo-SCT) have been found in that haplo-SCT results in lower relapse rate.
- We are interested in the mechanism of transplantation types on mortality.
- Sample size $n = 303$.

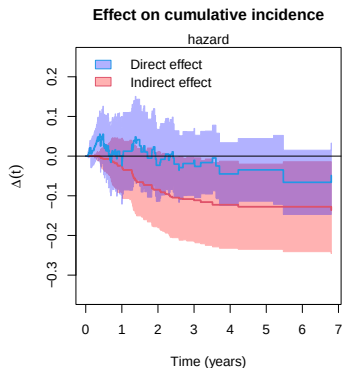
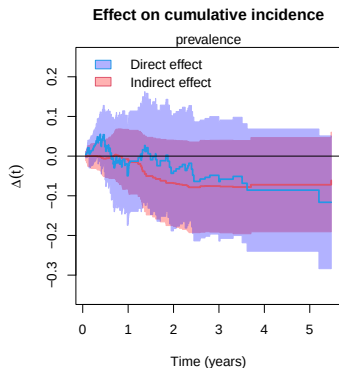
Estimating Counterfactual CIFs

- Different assumptions identify different $F(t; 0, 1)$.



Estimating Natural Effects

- Significant indirect effect: haplo-SCT reduces overall mortality through reducing the hazard of relapse.



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Summary

- We discussed the mediation analysis for time-to-event data, also known as semi-competing risks data.
- A significant challenge for time-to-event data is that the outcome variables are processes. The potential outcomes are defined on counting processes.
- The pointwise variance of estimators can be derived by the stochastic processes theory.

Extensions

- Time-varying confounders.
- Unobserved confounders.
- Pathway effects.
- Multiple intermediate events (multi-state data).

Reference

- Natural effects Deng Y, Wang Y, Zhou X-H. Direct and indirect effects in the presence of semi-competing risks. *Biometrics*. 2024.